KNOWLES & SKRODERMOOSE

THE ALL HALLOWS

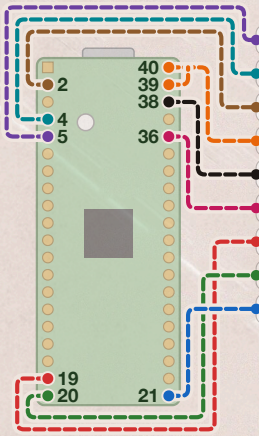
SPOOKY PUMPKIN EYES™

The WiFi Brain Transplant

1 **SWITCH OFF.** Switch the battery pack off (the switch is on its back).

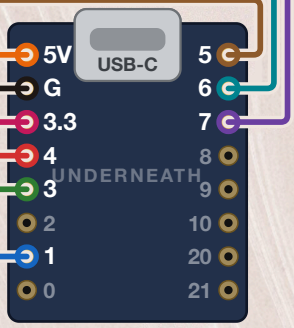
2 **MOVE THE WIRES.** Move the 9 wires to the new board **one at a time**: pull a wire off the Pico, check which pin it was on, push it onto the matching new pin. **Your wire colours may not match the drawing** (only red for + and black for – are usual), so go by the pins: if unsure, trace the wire back to the name printed on the eye board (DIN, CLK, CS). Keep the Pico: it still works.

OLD PICO, FROM ABOVE



WIRE	WAS ON THE PICO	NEW PIN
Eyes DIN	GP3 · pin 5	7
Eyes CLK	GP2 · pin 4	6
Eyes CS	GP1 · pin 2	5
+ Power	VBUS 40 or VSYS 39	5V
– Ground	GND · e.g. pin 38	G
Mouth long leg	3V3 · pin 36	3.3
Mouth red	GP14 · pin 19	4
Mouth green	GP15 · pin 20	3
Mouth blue	GP16 · pin 21	1

NEW BOARD, FROM UNDERNEATH



+ **Power** is the battery pack's + and the eyes' VCC;
– **Ground** is the battery pack's – and the eyes' GND.
Two separate + (or –) wires? Twist both onto the pin.
Match the pin names printed on the new board.

3 **SWITCH ON.** The eyes start their show straight away and, the first time, scroll the WiFi name, password and address for step 4. It's alive!

4 **INTRODUCE IT TO YOUR WIFI** (first time only). On a phone, join the WiFi called **Spooky Pumpkin**, password **spookyeyes**. If the phone says "no internet", choose *Use without internet* or *Stay connected*. A page pops up (if not within 20 seconds, open **192.168.4.1**): give your pumpkin a name (e.g. *Jack's Pumpkin*), pick your home WiFi, enter its password and tap **Save**. It only sees 2.4 GHz WiFi: pick your main network, not a guest one (if there are two, the one ending *2.4G*).

5 **PLAY.** It restarts, joins your home WiFi and scrolls its address, made from its name, e.g. **http://jacks-pumpkin.local** or numbers like **192.168.1.23**. Open it on a phone or computer on your WiFi for the button page: dozens of effects, a talking pumpkin (your phone speaks), your own words on the eyes, a page to make your own show, and Settings. If the name won't open (common on Android), use the numbers. Eyes say *Join WiFi* again? Probably a wrong password: join the pumpkin's WiFi and try again (the page says what went wrong).

ON THE DOORSTEP? If it loses your home WiFi for 3 minutes, it makes its own: join it and open **192.168.4.1**. It goes back to your WiFi by itself.

BOOT BUTTON: a quick press repeats the address, or the setup message (which stops after 15 minutes). Hold it for 5 seconds to forget your home WiFi.

HUNGRY BRAIN: fit fresh AA batteries, or plug a USB power bank into the USB-C socket (battery pack off).